CS641: Artificial Neural Network
University of Balochistan
Homework 5: Perceptron and Multilayer Network
Due date: 1, Nov 2024

July 2024
Computer Science and IT
Instructor: Dr. Abdul Basit

Fundamentals of Neural Network

Do the following

1. Do implement the perceptron algorithm with OR and AND dataset. Do also write why we cannot use perceptron for XOR dataset?
2. What is Backpropagation?
3. Write various phases of multilayer network that is forward pass and background pass.
4. Implement with pure python code the non linearly separable XOR and MNIST dataset with multilayer perceptron.
5. What are TensorFlow and Keras?
6. Create a separate virtual environment for TensorFlow Keras, and install tensorflow using the handouts.
7. Do implement MNIST full and CIFAR0-10 with TensorFlow keras in JupytorNotebook or Google Colab.

Note: Submit the assignment with printed screenshot captured for each question, Also print this assignment and attach with your assignment.